Chipsee

Industrial PC

# PPC-CM4-050




PN: CS12720RA4050P

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# PPC-CM4-050



Front View




Rear View




Side View 1




Side View 2

## 1 Product Overview

The Cortex®-A72 Raspberry Pi® series PPC-CM4-050 (PN: CS12720RA4050P) is a high-quality industrial Pi PC. This single board computer features a 5.0" 5-point capacitive touch screen with 1.0mm tempered glass, a resolution of 1280 x 720 pixels and a brightness of 400 cd/$m^2$. The built-in screen makes it a ready-to-use Raspberry Pi display, with no separate monitor needed.

Key Applications

- Human Machine Interface  HMI
- Process Control
- Process Monitoring
- HMI
- IIoT node
- Environmental Monitoring
- PLC
- Automotive applications
- ATM...

It is available as a device hosed in a casing with bezels, thus facilitating different installation options:

- Installation on an industrial cabinet
- Integration with the existing equipment

The PPC-CM4-050 industrial Pi PC is based around the powerful Raspberry Pi® Compute Module 4, powered by the Quad Cortex®-A72 processor with a processor speed of 1.5GHz.

# 2 Ordering Options

Chipsee products can be customized during the ordering process. The product will be shipped with the pre-installed factory defaults if no extra requirements are specified. The table in the Specifications section provides information about the default options bundled with the product.

> **Note**
>
> You can order PPC-CM4-050 from the official Chipsee Store or from your nearest distributor.

## 2.1 Pi® CM4 Module

The Pi® Compute Module 4 appears in different versions depending on the size of the DDR4 and eMMC.
The PPC-CM4-050 industrial Pi PC does not include the CM4 Raspberry Pi® module by default. If you would like to purchase it with a CM4, you can select it at the Chipsee store during the ordering process.

## 2.2 Operating System

This product comes with a pre-installed Raspberry Pi OS. Chipsee software engineers have created all the drivers, so every hardware feature is readily available for any standard development tool.

If your project requires a different OS, please **Contact us**, and we'll make a customized version that suits your needs.

# 3 Specifications

The PPC-CM4-050 industrial Pi PC offers a broad range of performance and connectivity options for scalable integration, providing expandability according to future needs. Some of the key features are listed in the table below.

**Key Features**

| | PPC-CM4-050 |
|---|---|
| **CPU** | Raspberry Pi® CM4; Quad Cortex-A72 at 1.5GHz |
| **Storage** | Support for 1 x TF Card[3] |
| **RAM** | 2/4/8 GB, Based on CM4 |
| **eMMC** | 16/32 GB, Based on CM4 |
| **Display Size** | 5.0" |
| **Display Type** | IPS LCD |

| PPC-CM4-050 | |
|---|---|
| **Display Resolution** | 1280 x 720 |
| **Brightness** | 400 cd/m$^2$ |
| **Touch Technology** | Capacitive |
| **Touch Point** | 5 |
| **Glass** | 1.0mm Tempered Glass |
| **USB** | 2 x USB 2.0 Host, 1 x Type-C |
| **LAN** | 1 x Giga LAN |
| **Audio** | 3.5mm Audio Out Connector, 2W Speaker Internal |
| **Buzzer** | Onboard Buzzer, driven by GPIO |
| **RTC** | High accuracy RTC with farad capacitor, can work 1 week after power off **(default)**. High accuracy RTC with lithium coin battery, can work 3 years after power off **(optional)**. |
| **RS232** | Default to 2 x RS232, up to 4 x RS232 |
| **RS485** | Default to 2 x RS485, these 2 x RS485 can be configured as 2 x RS232[1] |
| **CAN** | 1 x CAN-BUS |
| **GPIO** | Not Supported |
| **I2C** | Not Supported |
| **WiFi/BT** | Supported but depending on the CM4 selected[2] |
| **ZIGBEE** | No |
| **HDMI** | 1 x HDMI-D 2.0, can be driven upto 4Kp30 |
| **SATA II** | Not Supported |
| **3G/4G/LTE** | Not Supported |
| **Camera** | Yes, not mounted by default. Available on the board in the embedded PC. Requires a customized case to be exposed in an enclosed PC. |
| **Power Input** | From 6V to 36V |
| **Current** | 500mA Max at 12V |
| **Power Consumption** | 6W Typical |
| **Working Temperature** | From 0°C to +60°C |
| **OS** | Raspberry Pi OS |
| **Dimensions** | 139 x 85 x 27.1mm |
| **Weight** | 310g |
| **Mounting Method** | Panel, VESA |

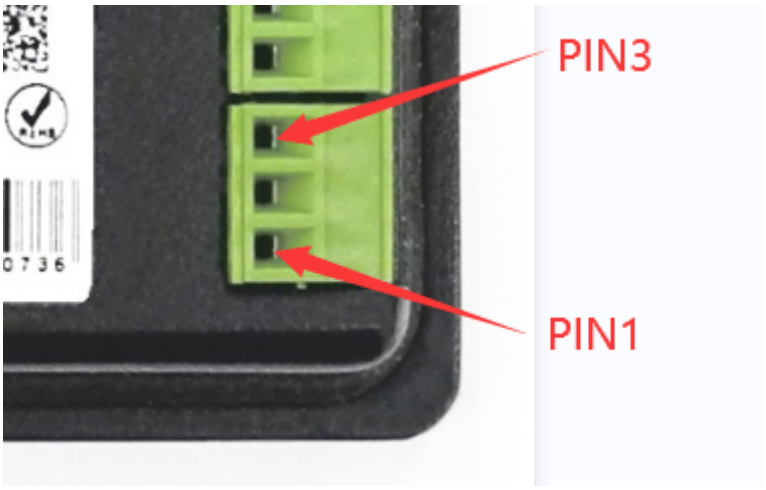
[1] The RS485 circuit controls the Input and Output direction automatically, there's no need to control it from within the software.

[2] The default product without the CM4 does not include a Wi-Fi/BT module. You can include a CM4 that has the Wi-Fi/BT module at the Chipsee store during the ordering process.

[3] Chipsee designed TF card slots for storage expansion, as the TF card for storage expansion use same pins with WiFi, it can't be used with WiFi at same time

# 4 Power Input

The PPC-CM4-050 industrial Pi PC can be powered by a wide range of input voltages: From 6V to 36V DC. It is a **3 Pin, 3.81mm screw terminal** connector. As shown in the figure below.




Power Input

Note that the "**+**" sign represents the positive power input, and it is printed both at the casing and as a silk-screen on the board of the embedded version. The "**-**" terminal is shorted to the ground.

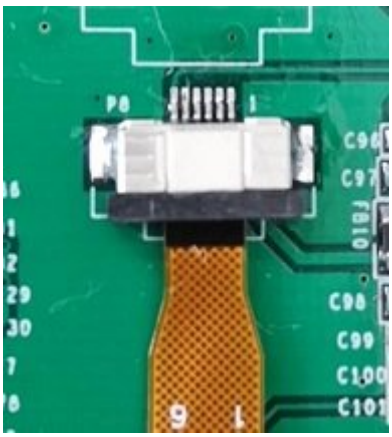
**Power Connector**

| **Power Input Definition** | | |
|---|---|---|
| **Pin Number** | **Definition** | **Description** |
| Pin 1 | Positive Input | DC Power **Positive Terminal** |
| Pin 2 | Negative Input | DC Power **Negative Terminal** |
| Pin 3 | Ground | **Power System Ground** |

> **Note**
>
> The system ground "**G**" is connected to power negative "**-**" on board.

# 5 Touch Screen

The PPC-CM4-050 industrial Pi PC uses a 5-point capacitive touch screen with 1.0mm tempered glass. However, the Raspberry Pi OS supports only One-Point touch.
The figure below shows the capacitive touch screen connected to the motherboard via the FPC connector.




Capacitive Touch Connector

**Attention**

A capacitive touch screen is susceptible to power noise and Electromagnetic Radiation (EMR). It may cause LCD ripples or even capacitive touch malfunction. If using a capacitive multi-touch test application, you might notice the touch points float erratically across the display. There are several solutions to this problem:

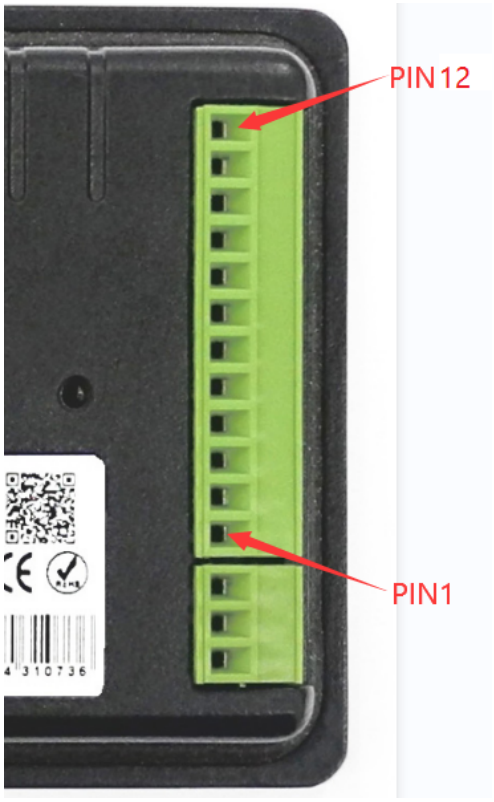
1. Use a high-quality Power Adapter Unit (PSU) with low EMR. You can also provide power from a battery.
2. Make sure that the PPC-CM4-050 Power Input connector (pin 3) is properly connected to the Power System Ground to provide sufficient EMI shielding and eliminate the problem entirely.
3. Bad GND problems can also be confirmed by touching pin 3 of the Power Input connector with one hand while operating the capacitive touch screen with the other hand. In this case, the operator's body acts as the Power System Ground.

# 6 Connectivity

There are many connectivity options available on the PPC-CM4-050 industrial Pi PC. It has 2 x USB 2.0 Host, 1 x Type-C, 1 x Giga LAN (RJ45) Ethernet connector supporting up to 1 Gbps, and 4 x UART and 1 x CAN terminals (RS232/RS485/CAN).

## 6.1 RS232/RS485/CAN

The serial communication interfaces (RS485, RS232, and CAN) are routed to a 12-pin 3.81mm terminal, as illustrated in the figure below.




RS232-RS485-CAN on the PPC-CM4-050 Industrial PC

**Attention**

1. RS485_3 and RS485_5 can control the input and output direction automatically. There's no need to control it from within the software.
2. The 120Ω match resistor for RS485 is **already** mounted by default.
3. The 120Ω match resistor for CAN is **NOT** mounted by default. Be sure to mount the match resistor when testing CAN.
4. The 2 x RS485 can be configured to 2 x RS232, if you want a custom configuration, you can contact us when placing an order.

The table below offers more detailed description of every pin and its definition:

**Connectivity Section**

| RS232 / RS485 / CAN Pin Definition: | | |
|---|---|---|
| **Pin Number** | **Definition** | **Description** |
| Pin 12 | CAN1_H | CAN H signal |
| Pin 11 | CAN1_L | CAN L signal |
| Pin 10 | RS485_4- | CPU UART5, RS485 –(B) signal |
| Pin 9 | RS485_4+ | CPU UART5, RS485 +(A) signal |
| Pin 8 | RS485_3- | CPU UART3, RS485 –(B) signal |
| Pin 7 | RS485_3+ | CPU UART3, RS485 +(A) signal |
| Pin 6 | RS232_0_RXD | CPU UART2, RS232 RXD signal |
| Pin 5 | RS232_0_TXD | CPU UART2, RS232 TXD signal |
| Pin 4 | RS232_2_RXD | CPU UART0, RS232 RXD signal, Debug Port |
| Pin 3 | RS232_2_TXD | CPU UART0, RS232 TXD signal Debug Port |
| Pin 2 | GND | System Ground |
| Pin 1 | +5V | System +5V Power Output, No more than 1A Current output |

## 6.2 USB Connectors

There are 2 x USB 2.0 Host, 1 x Type-C onboard, as shown in the figure below.



USB HOST Connectors

**Attention**

1. These two USB host connectors can drive 500mA for each channel at most.
2. These two USB host connectors come from the same USB HUB.
3. When you connect this product to the HOST PC through the Type-C port, the USB HUB will be disabled. As a result, the two USB host connectors will not work.

The product has one USB Type-C OTG connector that works as a slave by default. You can use it to establish a connection with the host PC and for downloading the system to the eMMC of CM4 module.



USB Type-C OTG Connector

**Warning**

1. Be careful not to touch surrounding electronic components accidentally while plugging in USB devices into the embedded Industrial PC version.
2. Remember to unplug the Type-C cable after flashing OS, otherwise the USB hosts won't work.

## 6.3 LAN

The 1 x Giga LAN provides Ethernet connectivity over standardized Ethernet cables as shown in the figure below. The integrated Ethernet interface supports up to 1 Gbps data throughput. These Giga LAN signals come from the CM4 module directly.



RJ45 LAN Connector

**Note**

Use CAT5 or better cables to achieve full data throughput over maximum distance defined by the 1000BASE-T standard (100m).

## 6.4 WiFi & BT Module

The default PPC-CM4-050 without the CM4 does not include a Wi-Fi/BT module.
If you include a CM4 that has the Wi-Fi/BT module, the product will have Wi-Fi/BT feature. The enclosed (CS12720RA4050P) variant of the product also includes an SMA connector for an external WiFi/BT antenna, as illustrated in the figure below.

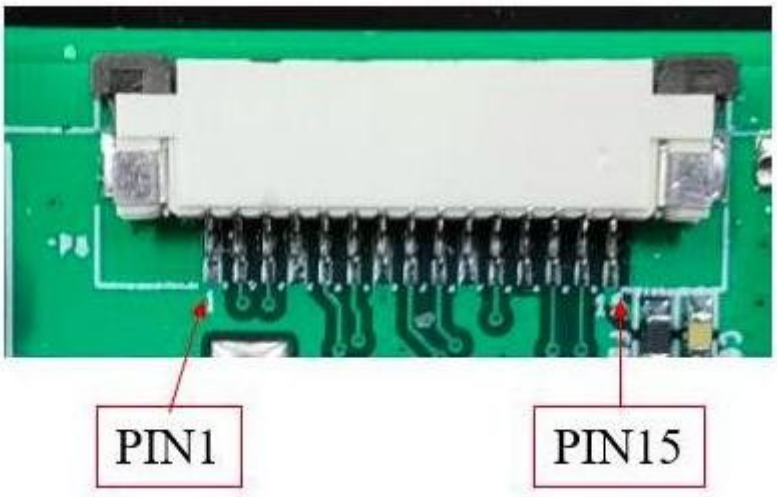



WiFi+BT Antenna

> **Attention**
>
> The product does not come shipped with the Wi-Fi/BT module by default.

## 6.5 Camera Connector

The PPC-CM4-050 industrial Pi PC has a 15 Pin **Camera Connector**, as shown in the figure below. The camera signals come from CAM1. The table below gives details about the definition of every Pin.




Camera Connector

**Camera Connector Pin-out**

| Camera Connector Pin Definition: | | |
|---|---|---|
| **Pin Number** | **Definition** | **Description** |
| Pin 1 | GND | Power Ground |
| Pin 2 | CAM1_DN0 | CAM1_DN0 |
| Pin 3 | CAM1_DP0 | CAM1_DP0 |
| Pin 4 | GND | Power Ground |
| Pin 5 | CAM1_DN1 | CAM1_DN1 |
| Pin 6 | CAM1_DP1 | CAM1_DP1 |
| Pin 7 | GND | Power Ground |
| Pin 8 | CAM1_CN | CAM1 Clock signal Negative |
| Pin 9 | CAM1_CP | CAM1 Clock signal Positive |
| Pin 10 | GND | Power Ground |
| Pin 11 | CAM GPIO | CAM GPIO, use for disable camera power and module |
| Pin 12 | NC | Not connected |
| Pin 13 | SCL0 | CPU I2C SCL0 signal |
| Pin 14 | SDA0 | CPU I2C SDA0 signal |
| Pin 15 | +3.3V | System +3.3V Power Output, No more than 500mA Current output |

**Attention**

1. The camera connector is supported but not mounted by default. It's available on the PCB but not exposed on the case, please contact us when placing an order if you need to use camera on the PPC-CM4-050.

# 7 TF Card Slot

The PPC-CM4-050 industrial Pi PC features 1 x **TF Card (micro SD) slot**. A slot can address up to 128GB of memory.



TF (micro SD) Card Slot

**Attention**

1. The SD is used for memory extension. It can't be used for system boot-up.
2. This storage extension SD uses the same pins as WiFi on CM4, SD storage and WiFi **can't** be used at the same time.
3. The product does not come shipped with the TF card by default.

## 8 Audio Connectors

The PPC-CM4-050 industrial Pi PC features some audio peripherals. It has 1 x **3.5mm audio output jack**.
Also, the PPC-CM4-050 industrial Pi PC has a miniature 2W internal speaker for audio reproduction, as well as a small buzzer for alarm/notification sounds.



Audio Connector

**Attention**

By plugging in the headphone cable, the internal speaker will be disabled automatically.

## 9 HDMI Connector

The PPC-CM4-050 industrial Pi PC supports 1 x HDMI-D 2.0 interfaces capable of driving 4K displays.



*HDMI Connector*

## 10 PROG Button

The PPC-CM4-050 industrial Pi PC has one button for entering usb download mode, as shown in the figure below.
When booting **with** the button being pressed, the Raspberry Pi will boot from the USB connector. You can use this feature to download the OS software to the internal eMMC.
When booting **without pressing** the button, the Raspberry Pi will boot from the internal eMMC.

There is no need to press the button during regular operation. However, if you need to reinstall the OS, please refer to the detailed information on how to reflash the OS from the Software Documentation.



PROG Button

# 11 Mounting Procedure

The PPC-CM4-050 industrial Pi PC can be mounted with 2 x M4 screws, enabling simplified installation onto any standard mounting fixture.

## 11.1 CS12720RA4050P

You can mount CS12720RA4050P with the Vesa (75 x 75mm) and Panel mounting methods, as shown in the figure below.




*Panel mounting*

> **Attention**
>
> Please make sure the display is not exposed to high pressure when mounting into an enclosure.

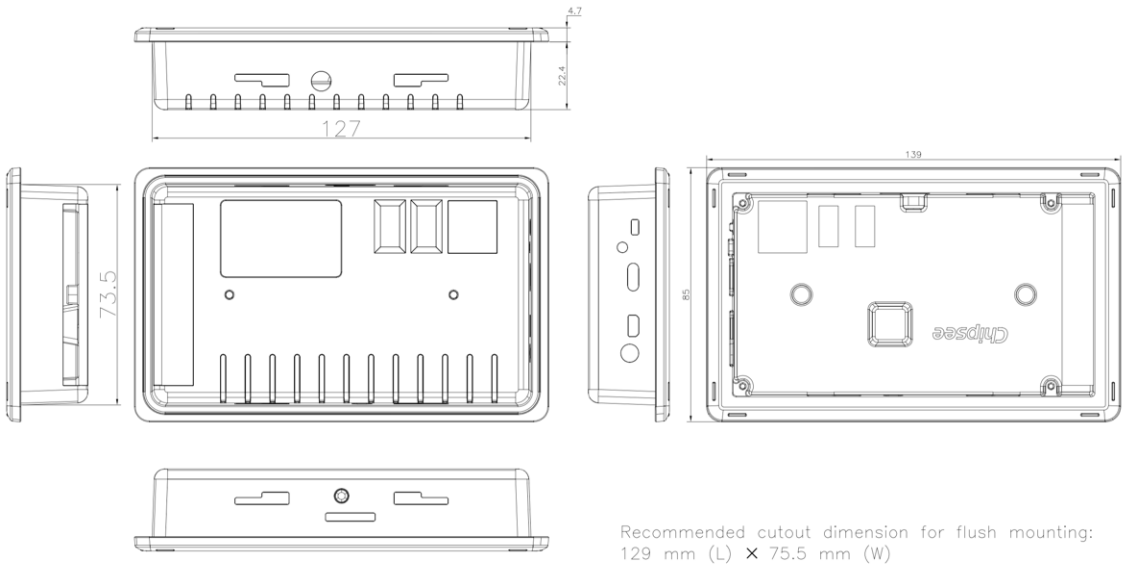
You can find detailed information about mounting in the Mount IPC Guide.

# 12 Mechanical Specifications

## 12.1 CS12720RA4050P

For CS12720RA4050P, the outer mechanical dimensions are 139 x 85 x 27.1mm (W x L x H).

Please refer to the technical drawing in the figure below for details related to the specific product measurements.




CS12720RA4050P *Technical Drawing*

## 13 3D Model

PPC-CM4-050 3D model can be viewed in the online doc in a web browser, **if you are reading from the PDF** version, please visit the online doc PPC-CM4-050, select hardware documentation, drag the navigation bar to the 3D Model section.

## 14 Disclaimer

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# 15 Technical Support

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